

Prompt

Role:

Act as a Senior QA Engineer with expertise in Banking, UPI, and Digital Payment Systems.

Scenario

A customer attempts to transfer ₹50,000.

The transaction fails.

The application displays the following generic error message:

"Transaction could not be completed."

Instructions

Analyze the scenario using step-by-step reasoning without assuming a single root cause.

Perform a comprehensive root cause analysis by identifying all possible reasons that could lead to the transaction failure.

For each possible reason:

1. Identify the root cause.
2. Explain why the failure could occur.
3. Categorize the root cause (e.g., Customer Validation, Account Validation, Business Rule, Security, Beneficiary, Network, Backend Service, Core Banking, External System, Fraud Detection, etc.).
4. Identify the system validation that detects the failure.
5. Explain the expected system behavior.
6. Mention the expected error message (if applicable).
7. Generate at least one positive test case.
8. Generate at least one negative test case.

Output Format

Provide the output in the following tables:

Table 1 – Root Cause Analysis

Category	Possible Root Cause	Reason	System Validation	Expected System Behaviour	Expected Error Message
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Table 2 – Positive Test Cases

| Test Case ID | Scenario | Preconditions | Test Steps | Expected Result | Priority |

Table 3 – Negative Test Cases

| Test Case ID | Scenario | Preconditions | Test Steps | Expected Result | Priority |

Table 4 – Summary

| Total Root Causes Identified | Functional Failures | Technical Failures | Security Failures | Infrastructure Failures | External Dependency Failures |

Output:

Positive Test Cases

TC ID	Scenario	Preconditions	Test Steps	Expected Result	Root Cause Covered
TC_POS_001	Transfer ₹50,000 successfully	Sufficient balance, daily limit available, beneficiary active	1. Login 2. Select beneficiary 3. Enter ₹50,000 4. Enter valid OTP 5. Click Transfer	Transaction succeeds and success message is displayed	Happy Path
TC_POS_002	Transfer within daily limit	Daily limit not exceeded	Transfer ₹50,000	Transaction completes successfully	Daily Limit Validation
TC_POS_003	Transfer with valid OTP	OTP is valid and not expired	Complete transfer using valid OTP	Transaction succeeds	OTP Validation
TC_POS_004	Transfer to active beneficiary	Beneficiary status is Active	Transfer funds	Transaction succeeds	Beneficiary Validation

TC ID	Scenario	Preconditions	Test Steps	Expected Result	Root Cause Covered
TC_POS_005	Transfer when all banking services are available	Bank server, Core Banking and UPI services are operational	Perform transfer	Money transferred successfully	Infrastructure

Negative Test Cases

TC ID	Scenario	Preconditions	Test Steps	Expected Result	Root Cause Covered
TC_NEG_001	Daily transfer limit exceeded	Customer has already reached daily limit	Attempt another ₹50,000 transfer	Transaction fails with appropriate message	Daily Limit Exceeded
TC_NEG_002	Insufficient account balance	Balance less than ₹50,000	Attempt transfer	Transaction rejected	Insufficient Balance
TC_NEG_003	OTP expired	OTP validity expired	Enter expired OTP	Transaction fails	OTP Expired
TC_NEG_004	Beneficiary inactive	Beneficiary account is inactive	Attempt transfer	Transaction rejected	Beneficiary Inactive
TC_NEG_005	Bank server unavailable	Banking server is down	Initiate transfer	Transaction fails gracefully	Bank Server Unavailable
TC_NEG_006	Network timeout during transfer	Network interruption occurs	Click Transfer	Appropriate timeout message displayed	Network Timeout
TC_NEG_007	Fraud rule triggered	Transaction exceeds fraud threshold	Initiate transfer	Transaction blocked for security	Fraud Detection

TC ID	Scenario	Preconditions	Test Steps	Expected Result	Root Cause Covered
TC_NEG_008	Core Banking service unavailable	CBS is down	Attempt transfer	Transaction fails without debit	Core Banking Unavailable
TC_NEG_009	UPI Switch unavailable	UPI switch is down	Initiate transfer	Transaction fails and user is informed	UPI Switch Unavailable